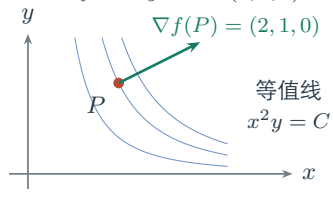
M1-12 本题的等值线、梯度与旋度方向

$z=0$：$f=x^2y$；$P=(1,1,0)$



$\mathbf{A}=(-y,x,0)$：绕 z 轴逆时针

$\nabla\cdot\mathbf{A}=0$；$\nabla\times\mathbf{A}=2\mathbf{e}_z$

$$f = x^2y$$
$$\mathbf{A} = (-y,x,0)$$

$$\nabla f = (2xy, x^2, 0)$$

$$\nabla\cdot\mathbf{A} = 0$$
$$\nabla\times\mathbf{A} = 2\mathbf{e}_z$$